

Quiz: Instruction formats

This quiz has 5 questions. The passing grade is 4. [Back](#) [Next](#)

Started on	Wednesday, January 15, 2025, 12:16 AM
State	Finished
Completed on	Wednesday, January 15, 2025, 12:20 AM
Time taken	3 mins 35 secs
Grade	1.00 out of 5.00 (20%)



Question **1**

Incorrect

0.00 points out of 1.00

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In EBCDIC, X'F0' to X'F9' represent the characters '0' to '9'. Which of these are equivalent as an immediate operand that specifies the EBCDIC character '1'? (Select 4)

Select one or more:

- ☒ A. C'F1' ✕
- ☐ B. 241
- ☐ C. B'11111111'
- ☐ D. B'11110001'
- ☒ E. X'F1' ✓
- ☒ F. X'01' ✕
- ☒ G. C'1' ✓

EBCDIC character '1' is:

C'1' (obviously!)

X'F1' (the hexadecimal representation of character '1' in EBCDIC)

B'11110001' (the binary representation of X'F1')

241 (without a prefix and quotes, this is decimal 241, or X'F1')

You can use any of the above to specify character '1', but C'1' is the preferred choice, as it makes the source program easier to understand.

The correct answers are: C'1', X'F1', B'11110001', 241

Question **2**

Incorrect

0.00 points out of 1.00

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The ADD REGISTER instruction has Opcode X'1A'. In memory, "Add register 4 to register 5" (in Assembler, "AR 5,4") will be

Select one:

- ☐ A. X'1A0405'
- ☐ B. X'451A'
- ☐ C. X'1A45'
- ☒ D. X'1A0504' ✖
- ☐ E. X'1A50004000'
- ☐ F. X'1A54'
- ☐ G. X'1A40005000'

The Assembler instruction 'AR 5,4' adds register 4 to register 5 and places the result on register 5. It becomes, in memory, X'1A54':

1A = Opcode

5 = First operand, register 5

4 = Second operand, register 4

X'1A45' would be 'AR 4,5' (it places the result on register 4).

X'451A' is an instruction with opcode X'45', which is not ADD.

The other options are longer than one halfword, so not valid RR-type instructions.

The correct answer is: X'1A54'

Question **3**

Correct

1.00 points out of 1.00

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If the index register of an RX-type second operand is zero, the effective address consists of base and displacement only

Select one:

☒ True ✓

☐ False

Correct!

In both base-displacement and indexed addressing, if the base or index registers are zero their contents are ignored when computing the effective address.

The correct answer is 'True'.

Question **4**

Incorrect

0.00 points out of 1.00

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In an RX-format instruction

Select one:

- ☐ A. One operand is a register and another a 16-bit relative address
- ☐ B. One operand is a register and another a base-displacement storage address
- ☒ C. Both operands are storage addresses ✕
- ☐ D. One operand is a storage address and another an immediate value
- ☐ E. One operand is a register and another an indexed storage address
- ☐ F. One operand is a register and another a long displacement storage address
- ☐ G. Both operands are registers

The correct answer is: One operand is a register and another an indexed storage address

Question 5

Incorrect

0.00 points out of 1.00

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In an RXY-format instruction

Select one:

- ☐ A. One operand is a register and another an indexed storage address with long displacement
- ☐ B. Both operands are storage addresses
- ☐ C. One operand is a register and another a base-displacement storage address
- ☒ D. Both operands are registers ✖
- ☐ E. One operand is a register and another an indexed storage address with 12-bit displacement
- ☐ F. One operand is a register and another a 16-bit relative address
- ☐ G. One operand is a storage address and another an immediate value

The correct answer is: One operand is a register and another an indexed storage address with long displacement